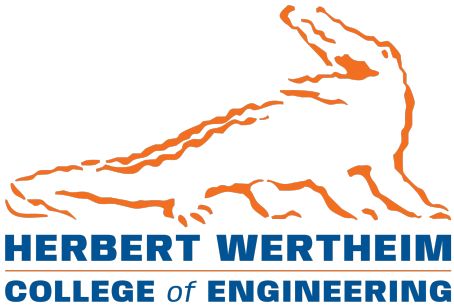
T2: Beta Test Plan

EEL4908 CpE Design 2

Hardened RISC-V Exam Computer

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1 Scope

Beta objective. Two boards accept a "start SEB" command from the server, and SEB launches on both.

Repository: `https://github.com/cannon-spencer/riscv-exam-computer`. Commit tested: `027d47894359fdc4bdf7c9e229fe81d56dff1a36`

In scope. Control server, board agent, start and stop scripts and sudoers rule, the install scripts, the SEB kiosk session on the board, and the lockdown and performance items carried over from alpha.

Out of scope. Canvas exam completion; kernel and boot-chain hardening (including `scripts/verity`); an instructor web interface (`exam-env/frontend` is empty, so the instructor uses `curl`); server-pushed exam configuration; and the separate GPU/cage build, which lives on one isolated board and is not in the repository.

Notation. A value marked † is a proposed target chosen by the team, not derived from the code or a stakeholder requirement. The first full test run records a baseline, and the team then confirms or revises each such value.

2 Alpha test results

Results come from the M3 alpha report and the dated engineering logs in `docs/log/`. Where no result was recorded, the table says so.

2.1 Results against the T1 plan

T1 ID	Result	Evidence and notes
HW-01	Pass	Image flashes and boots to a login over UART (M3 report).
HW-02	Pass	SSH works on a lab LAN. On campus Wi-Fi the board must be registered as a UF device and reached over Tailscale (log 2026-09-23).
HW-03	Pass	Outbound IP works over Wi-Fi (M3 report).
SW-01	Pass, approach changed	Building `seb-linux` on each board failed repeatedly (log 2026-09-23). SEB is now built once on a board and published as a riscv64 release binary that `install-seb.sh` copies to each board (log 2026-09-25).
INT-01	Pass, with a performance caveat	SEB launches in an `xinit` session with `openbox` on virtual terminal 3 and loads Canvas, but clicks take 10 s or more (log 2026-09-30).
INT-02	Not run	No lockdown result is recorded. The test is carried into beta as LD-01.
SW-02	Pass	Heartbeat reached a LAN listener (M3 report) and later the Cloudflare-tunnelled server (log 2026-09-23).

Table 1: Alpha results.

2.2 How alpha testing changed the project

- **On-board SEB builds were unreliable** (log 2026-09-23). SEB is now built once on a board and installed from a riscv64 release binary.
- **Campus networks block inbound connections.** The control server moved from an Oracle Cloud VM, which ran out of memory on package installs, to a laptop behind a Cloudflare tunnel, so boards only make outbound HTTPS requests (log 2026-09-23). Eduroam also blocks SSH to the board, so boards are registered as UF devices and reached over Tailscale.
- **Canvas is very slow on the board.** Clicks take 10 s or more and the page paints in tiles (log 2026-09-30). A separate investigation on an isolated board found the SEB web process at 100% of one core and a post-login load of over two minutes in a single timed run. That work is not in the repository.
- **Thermals and display glitches.** Two sessions of about 12 hours ran without a crash, but the board reached about 75 °C with no heatsink or fan, and the display glitched several times (log 2026-09-30).
- **A failed UF login leaves the user stuck** on `ufl.edu` under SEB's navigation lockdown (log 2026-09-30). It is recorded as BUG-03 in the M4 report and has no beta test.
- **Control grew from a heartbeat to a job queue.** The server now keeps an inventory and one pending job per board, and the agent starts and stops SEB through sudo-whitelisted scripts (log 2026-09-30).

2.3 How alpha testing shaped this plan

- Release-binary install path: INS-01.
- Network and control path: SRV-01 to SRV-09, AGT-02 to AGT-05, SEC-02, SEC-03.
- Canvas speed: PERF-01, PERF-02.
- Thermals and glitches: PERF-03, PERF-04.
- Lockdown was not tested in alpha: LD-01.
- Job queue: SRV, AGT, SES and DEMO tests, and EDGE-01.
- The server identifies boards by hostname, so two boards flashed from one image can collide: AGT-01.
- Failed-login trap: no beta test; tracked as BUG-03.

3 Expected behavior

This section describes behavior as implemented in the source at the tested commit.

3.1 System overview

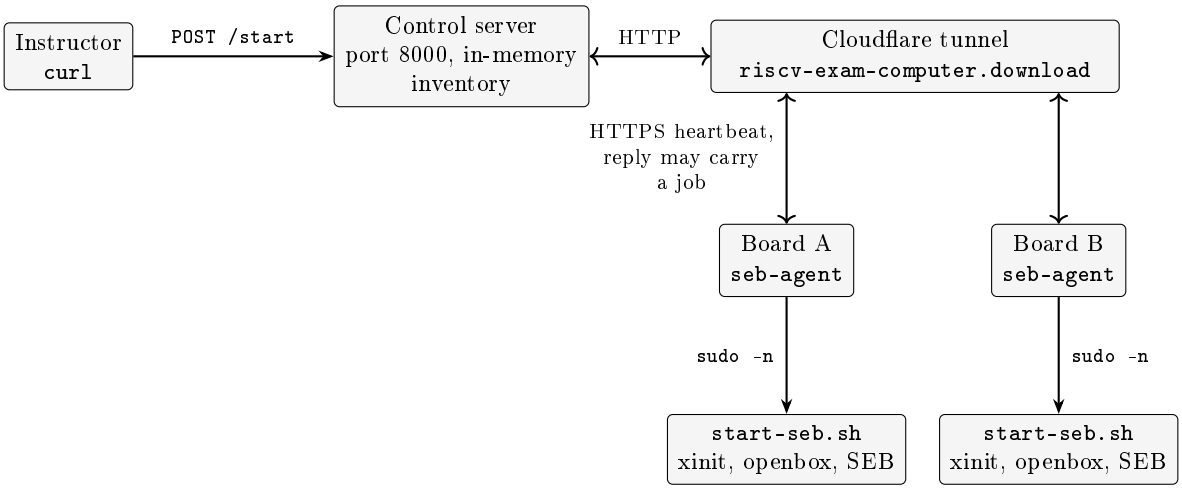


Figure 1: Control path. Boards poll the server; the server never connects to a board.

3.2 Control server

The server (`exam-env/server/src/main.rs`) listens on port 8000 and holds, per host name, the last heartbeat time, the last reported state and at most one pending job (`start` or `stop`).

Route	Expected behavior
`POST /heartbeat`	Body `{"host":...,"state":...}`. Records the host, time and state. Replies `{"ok":true,"job":...}` where `job` is `"start"`, `"stop"` or `null`. Missing `host`: 422. Invalid JSON: 400. No JSON content type: 415.
`GET /machines`	Lists every host seen since the server started, with `state`, `pending` and `seen_secs_ago`, sorted by host.
`POST /start`	Sets pending `start` on every host whose last heartbeat is 15 s old or less, and returns the sorted list `queued`. Returns `"queued":[]` when no host is online.
`POST /stop`	Same as `/start` with job `stop`.

Table 2: Server routes.

Job delivery rules (checked by SRV-01 to SRV-09):

- A job is handed out in the reply to the board's next heartbeat and then cleared.
- A pending `start` is not delivered while the board reports `running`; it stays pending and is delivered the first time the board reports another state.
- A pending `stop` is delivered regardless of reported state.
- There is one pending slot per board, so a later command replaces an earlier undelivered one.
- Inventory is held in memory only.

3.3 Board agent

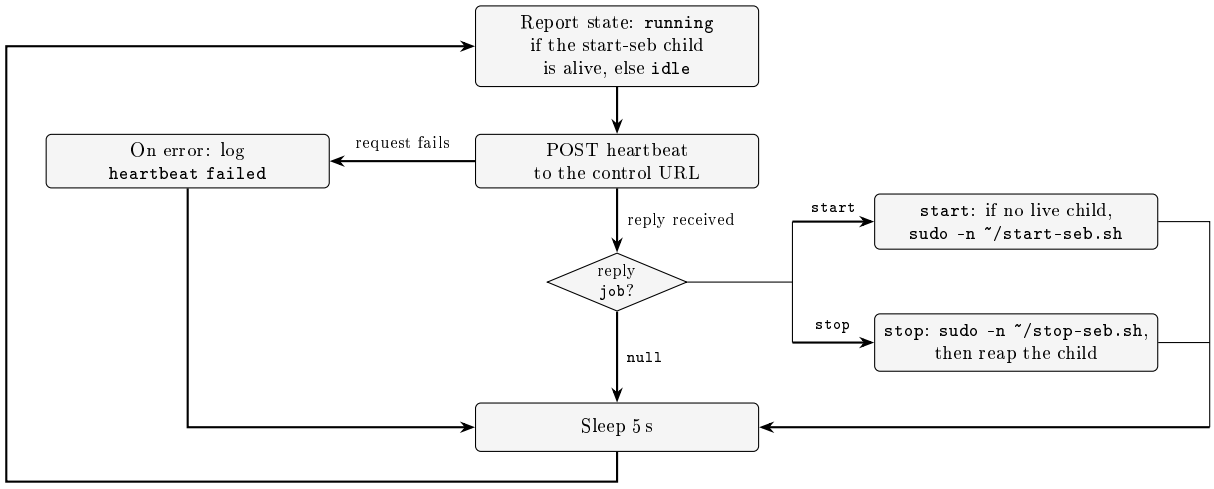


Figure 2: Agent loop (`exam-env/seb-agent/src/main.rs`).

- The agent sleeps 5 s after each POST, so the observed period is 5 s plus the request. It reports `running` while the `sudo`/`xinit` child from `start-seb.sh` is alive, otherwise `idle`.
- The reported host name is the `HOSTNAME` environment variable, set by the systemd unit from the `%H` specifier.
- The control URL is compiled into the agent: `https://riscv-exam-computer.download/heartbeat`.
- The agent opens no listening port.
- Sudo for the agent is limited to `/home/orangepi/start-seb.sh` and `/home/orangepi/stop-seb.sh`.

3.4 Board session

Start. `start-seb.sh` runs as root and execs `xinit` on display `:1`, virtual terminal 3, starting `openbox` and then `/home/orangepi/safe-exam-browser--windowed` with a local start file (paths hardcoded for the `orangepi` account).

Stop. `stop-seb.sh` kills the SEB and `xinit` processes and X1 socket holders, rebinds the framebuffer console, restores keyboard mode and switches to virtual terminal 1.

3.5 Quantified requirements

ID	Requirement	Value
R-T1	Heartbeat sleep after each request	5 s (observed period is sleep + RTT)
R-T2	A board counts as online while its last heartbeat is no older than	15 s (code)
R-T3	Time from `POST /start` to SEB running on every online board, in seconds	10 s†
R-T4	Maximum difference between the two boards' start times, in seconds	5 s†
R-T5	Time from `POST /stop` to console restored, in seconds	10 s†
R-P1	Canvas page load time, in seconds	login page $\leq$ 30 s, post-login page $\leq$ 60 s†
R-P2	Click-to-response latency in Canvas, in seconds	median $\leq$ 2 s†
R-P3	Maximum board temperature during a session, in °C	$<$ 80 °C†

Table 3: Quantified requirements. Values marked † are proposed targets.

3.6 Not implemented in the tested commit

- The server code has no authentication on any route.
- `/start` and `/stop` address all online boards; there is no per-board targeting.
- The server does not distribute exam configuration; a later change is a pushed `.seb`.
- Secure start (the exam opening only under SEB): no mechanism exists yet; see SEC-01.
- Paths and the account name are hardcoded (`orangepi`); the install scripts do not install `openbox`, `xorg` or `xinit`; and the local start file is not in the repository.

4 Test procedures

4.1 Equipment and conventions

Two Orange Pi RV boards, each with an HDMI display, a USB keyboard, power and a UART adapter (115200 baud); a laptop running Docker and `cloudflared` with the `exam-server` tunnel; a network that reaches `https://riscv-exam-computer.download` (a lab network, or `ufdevice` with Tailscale for registered boards); and a stopwatch or screen recording. Record the commit and board names for every run in Table 10.

4.2 Server (no hardware)

Start a local server on `:8000` (`cargo run` in `exam-env/server`, or Docker without the tunnel). Do not use `./scripts/deploy-server.sh` or the public URL for these tests: that path is the live tunnel, and live boards would consume jobs. Drive SRV-01 to SRV-09 with `curl` to `http://127.0.0.1:8000`. SRV-09 needs a 16 s wait after the last heartbeat.

ID	Procedure and pass criterion
SRV-01	**Heartbeat.** POST a heartbeat for a new host. **Pass:** reply contains `"ok":true` and `"job":null`.

ID	Procedure and pass criterion
SRV-02	**Inventory.** Heartbeat two hosts, then `GET /machines`. **Pass:** both hosts listed with state and `seen_secs_ago`.
SRV-03	**Broadcast start.** `POST /start` with both hosts online. **Pass:** `queued` lists both; the next heartbeat from a host returns `start`, the one after returns `null`.
SRV-04	**Start withheld while running.** Heartbeat with state `running` while `start` is pending. **Pass:** no job returned and `/machines` still shows `pending:"start"`.
SRV-05	**Stop always delivered.** `POST /stop`, then heartbeat from a running and an idle host. **Pass:** both receive `stop`.
SRV-06	**Single slot.** `/start` then `/stop` before the board polls. **Pass:** only `stop` is delivered.
SRV-07	**Pending start after exit.** `/start` while running, then report `idle`. **Pass:** `start` is delivered.
SRV-08	**Bad input.** Missing host, invalid JSON, no content type, `GET /start`. **Pass:** 422, 400, 415, 405, and `/machines` still answers 200.
SRV-09	**Offline exclusion.** Stop heartbeating for 16 s, then `POST /start`. **Pass:** `"queued":[]`.

Table 4: Server tests.

4.3 Agent and board

ID	Procedure and pass criterion
INS-01	**Fresh install.** Flash a card, run `install-software.sh` and `install-seb.sh` with `--host`, then any manual steps. Record every manual step needed. **Pass:** `systemctl --user is-active seb-agent` prints `active`, `~/safe-exam-browser` exists and is executable, and the board appears in `/machines`.
AGT-01	**Unique identity.** On each board run `hostname` and `systemctl --user show seb-agent -p Environment`. **Pass:** `/machines` lists each board as a separate host.
AGT-02	**Cadence.** `journalctl --user -u seb-agent -o short-precise -n 25`. **Pass:** intervals match R-T1; acceptable range 4.5 to 7 s.
AGT-03	**Outage tolerance.** Make the control URL unreachable for a set time, then restore it. **Pass:** the agent keeps running, logs each failed heartbeat, and resumes heartbeating after restoration; outage of 60 s†, with recovery within 15 s of restoration†.
AGT-04	**Agent crash recovery.** Kill the agent process. **Pass:** the systemd unit restarts it within 15 s†.
AGT-05	**Boot without login.** Power-cycle a board and do not log in. **Pass:** the board is listed in `/machines` within 90 s† of power-on.
SUDO-01	**Sudo scope.** As `orangepi` run `sudo -n -l`. **Pass:** only the two scripts are listed.
SES-01	**Start from console.** Run `sudo -n ~/start-seb.sh`. **Pass:** SEB is visible on the display within 30 s†.
SES-02	**Stop from console.** Run `sudo -n ~/stop-seb.sh`. **Pass:** no SEB or `xinit` process remains, the text console is shown and the keyboard works.
SES-03	**Repeated cycles.** Start and stop repeatedly, recording memory use each cycle. **Pass:** every cycle meets SES-01 and SES-02; 10 cycles†; available memory after the last cycle is within 10 %† of the first.

ID	Procedure and pass criterion
SES-04	**Duplicate start.** While running, send `POST /start` twice. **Pass:** a single SEB session.
SES-05	**Config.** Confirm the board has a local start file for the lab launch (not a repo artifact). **Pass:** SEB starts without a password prompt; the start page is the UF Canvas site.

Table 5: Agent and board session tests.

4.4 Integration and demo

ID	Procedure and pass criterion
DEMO-01	**Two boards start and stop on one command.** Pre: SES-01, SES-02 and AGT-01 passed on both boards; both listed `idle` in `/machines`. Steps: (1) note the time; (2) `curl -sS -X POST $CTRL/start`; (3) record when SEB appears on each display; (4) `GET /machines`; (5) `curl -sS -X POST $CTRL/stop`; (6) record when each display returns to the console. Three trials. **Pass:** `queued` lists both boards; SEB runs on both; `/machines` shows both `running`; after stop both are back at the console. Time limits are R-T3 to R-T5.
DEMO-02	**Board offline at start.** Stop one board's agent, send `POST /start`. **Pass:** `queued` lists only the other board. Restart the agent; record whether it launches SEB. Send `/start` again and record the result.
DEMO-03	**Board loses network while running.** With both running, disconnect one board. Record SEB state, the board's `seen_secs_ago`, and the `queued` list for a `POST /stop`. Reconnect and send `/stop` again.
DEMO-04	**Server restart.** With both boards running, restart the server. Record how long until both are listed again and with which state.
EDGE-01	**Pending start after self-exit.** While a board reports `running`, send `/start` (held), then tear the session down so the agent reports `idle`. **Pass:** the next heartbeat receives `start` and the session relaunches (job rule above).

Table 6: Integration and demo tests.

4.5 Lockdown and security

ID	Procedure and pass criterion
LD-01	**Exit paths.** With SEB started by the agent, try Alt+F4, Alt+Tab, Ctrl+Alt+F1 to F7, Ctrl+Alt+Backspace, Ctrl+Alt+Del, the Super key, Esc, right-click and middle-click on the desktop background, and moving, resizing or minimising the SEB window. **Pass:** none gives a shell, another application or the console. Record each attempt and outcome.
SEC-01	**Secure start.** Deferred. The intended behavior is that an exam can only be started from SEB and a normal browser is refused; no mechanism has been chosen, so there is nothing to test yet.

ID	Procedure and pass criterion
SEC-02	**No inbound surface.** On a board run `ss -ltnp`. **Pass:** no listening socket belongs to `seb-agent`.
SEC-03	**Control-channel access.** Record whether `POST /start` and `POST /stop` succeed from a machine with no credentials, with no exam running.

Table 7: Lockdown and security tests.

4.6 Performance and endurance

ID	Procedure and pass criterion
PERF-01	**Page load.** Start SEB with `/start`; time the Canvas login page and the post-login page. Five runs†. Report median and maximum. **Pass:** R-P1.
PERF-02	**Input latency.** Time click to visible response over ten clicks†. **Pass:** R-P2.
PERF-03	**Thermal.** Run a Canvas session of 60 minutes† while logging temperature once a minute by reading `/sys/class/thermal/thermal_zone*/temp`. **Pass:** R-P3.
PERF-04	**Endurance.** Run SEB for 3 hours†. Count crashes, agent restarts and display glitches. **Pass:** zero crashes, zero agent restarts and no heartbeat gap over 15 s; record every display glitch.

Table 8: Performance tests.

4.7 Demo runbook (DEMO-01)

```
# laptop: start server and tunnel (Docker running)
./scripts/deploy-server.sh

# second terminal
CTRL=https://riscv-exam-computer.download
curl -sS $CTRL/machines # both boards listed, state idle
curl -sS -X POST $CTRL/start # queued lists both boards
curl -sS $CTRL/machines # both boards state running
curl -sS -X POST $CTRL/stop # both boards back at the console
```

5 Traceability

Expected behavior	Tests
Server routes and job rules	SRV-01 to SRV-09, EDGE-01
Agent loop, identity, recovery	AGT-01 to AGT-05, SUDO-01, INS-01
Board session start and stop	SES-01 to SES-05
Two-board start and stop (beta objective)	DEMO-01 to DEMO-04
Timing requirements	AGT-02, DEMO-01, SES-01, SES-02
Performance requirements	PERF-01 to PERF-04
Lockdown and security	LD-01, SEC-01 to SEC-03
Alpha findings	INS-01, AGT-01, LD-01, SEC-02, SEC-03, PERF-01 to PERF-04

Table 9: Expected behavior to tests.

6 Results record

Test	Date	Tester	Boards	Result	Measurements and notes

Table 10: Results record template.